

Ejercicios de Nmap (Redes y Seguridad Informática)

1. Comprobar conectividad básica y escaneo simple

`nmap 192.168.1.143`

- Escanea el host con IP 192.168.1.143 para detectar puertos abiertos comunes.

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.113
```

2. Escaneo de un rango de IP

`nmap 192.168.1.1-254`

- Escanea todas las direcciones IP del rango 192.168.1.1 a 192.168.1.254.

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.1-254  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:07 EST  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 252 undergoing ARP Ping Scan  
ARP Ping Scan Timing: About 1.98% done; ETC: 04:07 (0:00:00 remaining)  
Nmap scan report for 192.168.0.110  
Host is up (0.0000020s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
  
Nmap scan report for 192.168.0.250  
Host is up (0.0000020s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh
```

3. Escaneo de varias direcciones IP específicas

`nmap 192.168.1.1 192.168.1.4 192.168.1.43`

- Escanea múltiples IP individuales especificadas manualmente.

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.113 192.168.0.130
```

4. Escaneo de puertos específicos

`nmap -p 1042 192.168.1.143`

- Escanea solo el puerto 1042 en la IP dada.

```
(kali㉿kali)-[~]  
$ nmap -p 38 192.168.0.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:08 EST  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 1.47 seconds
```

5. Escaneo de puertos UDP

`nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168.1.143`

- Escaneo en modo UDP de puertos específicos.
- El parámetro `-sU` indica que se trata de un escaneo de puertos UDP.

```
(kali㉿kali)-[~]  
$ nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168.0.130
```

6. Escaneo del sistema operativo

`nmap -O -osscan-guess localhost`

- Detecta el sistema operativo del host local.
- El parámetro `--osscan-guess` permite adivinar el sistema si no se reconoce con certeza.

```
(kali㉿kali)-[~]  
$ nmap -O -osscan-guess localhost  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:11 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000063s latency).  
Other addresses for localhost (not scanned): ::1  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
Device type: general purpose  
Running: Linux 2.6.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6  
OS details: Linux 2.6.32, Linux 5.0 - 6.2  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.46 seconds
```

7. Mostrar versión de los servicios

`nmap -sV -version-all localhost`

- Muestra la versión de los servicios detectados en el host local.

```
(kali㉿kali)-[~]  
$ nmap -sV -version-all localhost  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:11 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000030s latency).  
Other addresses for localhost (not scanned): ::1  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 8 (protocol 2.0)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds
```

8. Mostrar información del sistema Nmap instalado

`nmap -SV --version-1ll localhost`

- Posible errata tipográfica, probablemente se refiere a:

`nmap -sV --version-intensity 9 localhost`

```
(kali㉿kali)-[~]  
$ nmap -sV --version-intensity 9 localhost  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:24 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000030s latency).  
Other addresses for localhost (not scanned): ::1  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 8 (protocol 2.0)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds
```

9. Ver interfaces de red

`nmap --iflist localhost`

- Muestra todas las interfaces de red y rutas de la máquina local.

```
(kali㉿kali)-[~]
$ nmap --iflist localhost

Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:25 EST
*****INTERFACES*****
DEV (SHORT) IP/MASK                                TYPE    UP MTU    MAC
lo (lo)      127.0.0.1/8                                     loopback up 65536
lo (lo)      ::1/128                                       loopback up 65536
eth0 (eth0)  192.168.0.110/23                                  ethernet up 1500  08:00:27:D1:F8:5D
eth0 (eth0)  fe80::e91:d06b:115:7c7a/64                         ethernet up 1500  08:00:27:D1:F8:5D
eth1 (eth1)  10.0.3.15/24                                       ethernet up 1500  08:00:27:14:CB:1A
eth1 (eth1)  fd17:625c:f037:3:79b1:d9f:7503:64ce/64             ethernet up 1500  08:00:27:14:CB:1A
eth1 (eth1)  fe80::6fc8:b0b0:e488:cf1e/64                       ethernet up 1500  08:00:27:14:CB:1A

*****ROUTES*****
DST/MASK                                DEV  METRIC GATEWAY
10.0.3.0/24                             eth1  101
192.168.0.0/23                          eth0  100
0.0.0.0/0                               eth0  100    192.168.0.1
0.0.0.0/0                               eth1  101    10.0.3.2
::1/128                                 lo    0
fd17:625c:f037:3:79b1:d9f:7503:64ce/128 eth1  0
fe80::e91:d06b:115:7c7a/128            eth0  0
fe80::6fc8:b0b0:e488:cf1e/128          eth1  0
fd17:625c:f037:3::/64                  eth1  101
fe80::/64                              eth0  1024
fe80::/64                              eth1  1024
ff00::/8                                eth0  256
ff00::/8                                eth1  256
::/0                                    eth1  101    fe80::2
```

10. Enviar paquetes TCP ACK

`nmap -sA 192.168.1.143`

- Escanea con paquetes TCP ACK para detectar hosts activos incluso tras firewall.

```
(kali㉿kali)-[~]
$ nmap -sA 192.168.0.130

Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:26 EST
Nmap scan report for 192.168.0.130
Host is up (0.00047s latency).
All 1000 scanned ports on 192.168.0.130 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:FC:60:1C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

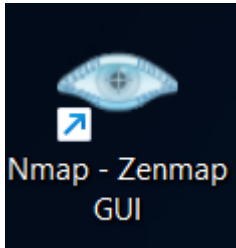
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

11. Actividad propuesta 1.9: Zenmap GUI para Nmap

🔧 Instrucciones del ejercicio:

Instala **Zenmap** (entorno gráfico para Nmap) en tu máquina Windows.

Investiga y explica **la utilidad de las diferentes pestañas** que ofrece la aplicación.



12. Otros comandos avanzados (mencionados en la imagen):

- Detección de DNS inverso:

`nmap -sL 192.168.43.0/24`

```
(kali㉿kali)-[~]  
$ nmap -sL 192.168.43.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:27 EST  
Nmap scan report for 192.168.43.0  
Nmap scan report for 192.168.43.1  
Nmap scan report for 192.168.43.2  
Nmap scan report for 192.168.43.3  
Nmap scan report for 192.168.43.4  
Nmap scan report for 192.168.43.5  
Nmap scan report for 192.168.43.6
```

- Escaneo de versión con `--version-trace`
- Exclusión de host:

`nmap 192.168.1.0/24 --exclude 192.168.1.1`

```
(kali㉿kali)-[~]  
$ nmap 192.168.0.0/24 --exclude 192.168.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-23 04:27 EST  
Stats: 0:03:05 elapsed; 200 hosts completed (54 up), 54 undergoing S
```